

MM-GBSA Binding Free Energy

Goal: Calculate the binding free energy (ΔG) between mexiletine and the Nav1.5 channel using the MM-GBSA method via gmx_MMPBSA.

Prerequisites: AmberTools (cpptraj, parmed, pdb4amber), GROMACS, gmx_MMPBSA. All available in the mmpbsa conda environment.

```
conda activate mmpbsa
mkdir -p ~/PHD_MD/MMPBSA_[SYS_NAME]
cd ~/PHD_MD/MMPBSA_[SYS_NAME]
```

PHASE 1: Trajectory Preparation (cpptraj)

Strip water, ions, and lipids from the trajectory and save a reference PDB and a stripped XTC file for gmx_MMPBSA.

Note: The residues stripped here must exactly match those stripped in the ParmEd step below.

```
SIM="/path/to/charmm-gui-[JOB_ID]"

cpptraj -p ${SIM}/openmm/step5_input.psf << EOF
trajin ${SIM}/openmm/step7_1.dcd
trajin ${SIM}/openmm/step7_2.dcd
trajin ${SIM}/openmm/step7_3.dcd
trajin ${SIM}/openmm/step7_4.dcd
trajin ${SIM}/openmm/step7_5.dcd
trajin ${SIM}/openmm/step7_6.dcd
trajin ${SIM}/openmm/step7_7.dcd
trajin ${SIM}/openmm/step7_8.dcd
trajin ${SIM}/openmm/step7_9.dcd
trajin ${SIM}/openmm/step7_10.dcd
strip :TIP3,OPC,POT,CLA,SOD,CHL1,POPC
trajout reference.pdb onlyframes 1
trajout stripped_traj.xtc offset 10
run
exit
EOF
```

Output:

- reference.pdb — single frame, stripped (used as structure reference)
- stripped_traj.xtc — every 10th frame, stripped (1,000 frames total)

⚠ Always strip OPC (not just TIP3) — this project uses OPC water model. Missing OPC causes atom count mismatch.

PHASE 2: Topology Conversion (ParmEd)

Convert the CHARMM PSF topology to GROMACS format for gmx_MMPBSA.

```
python << EOF
import parmed as pmd

SIM = "/path/to/charmm-gui-[JOB_ID]"

psf = pmd.load_file(f"{SIM}/openmm/step5_input.psf")
psf.coordinates = pmd.load_file(f"{SIM}/openmm/step5_input.crd").coordinates
```

```

psf.strip(":TIP3,OPC,POT,CLA,SOD,CHL1,POPC")

params = pmd.charmm.CharmmParameterSet(
    f"{SIM}/toppar/par_all36m_prot.prm",
    f"{SIM}/toppar/par_all36_lipid.prm",
    f"{SIM}/toppar/par_all36_cgenff.prm",
    f"{SIM}/toppar/par_all36_carb.prm",
    f"{SIM}/toppar/par_all36_na.prm",
    f"{SIM}/toppar/par_interface.prm",
    f"{SIM}/toppar/toppar_water_ions.str",
    f"{SIM}/unl/unl.prm"
)
psf.load_parameters(params)

psf.save("system.top")
psf.save("system_amber.pdb")
print("ParmEd done:", len(psf.atoms), "atoms")
EOF

```

Note: The SOD atom type warning from ParmEd is harmless and can be ignored.

PHASE 3: Fix PDB for gmx_MMPBSA (pdb4amber)

pdb4amber assigns chain IDs and fixes residue numbering issues that cause gmx_MMPBSA to fail:

```
pdb4amber -i system_amber.pdb -o system_final.pdb
```

Verify atom counts are consistent before proceeding:

```

grep -c '^ATOM\|^HETATM' system_final.pdb
grep -c '^ATOM\|^HETATM' reference.pdb

```

⚠ Both counts must match exactly. If they differ, the stripping step in cpptraj and ParmEd are inconsistent — check that the same residues are stripped in both.

PHASE 4: Create Index File

```
gmx make_ndx -f system_final.pdb -o index.ndx
```

Type q and press Enter to save and quit. Note the group numbers for Protein and UNL from the output. Typically:

- Group 1 = Protein
- Group 13 = UNL (ligand)

Note: Confirm the group numbers from the gmx make_ndx output before running gmx_MMPBSA.

PHASE 5: Create mmpbsa.in Input File

For trajectory-averaged MM-GBSA (recommended, more reliable):

```

cat > mmpbsa.in << 'EOF'
&general
sys_name="[SYS_NAME]",
startframe=1,
endframe=1000,
interval=5,
PBRadii=2,
temperature=310,
/

```

```
&gb
igb=5,
saltcon=0.150,
/
EOF
```

Note: PBRadii=2 (mbondi2) is required with igb=5 (OBC2 GB model). Using CHARMM default radii with igb=5 causes errors. startframe=1, endframe=1000, interval=5 analyzes 200 frames from the 1,000-frame stripped trajectory.

PHASE 6: Run gmx_MMPBSA

```
gmx_MMPBSA -O \
-i mmpbsa.in \
-cs system_final.pdb \
-ct stripped_traj.xtc \
-ci index.ndx \
-cg 1 13 \
-cp system.top \
-o FINAL_RESULTS_[SYS_NAME].dat \
-eo FINAL_RESULTS_[SYS_NAME].csv \
-nogui
```

Replace 1 and 13 with your actual group numbers from Step 4.

PHASE 7: Read Results

The main result is in the .dat file:

```
grep 'DELTA TOTAL' FINAL_RESULTS_[SYS_NAME].dat
```

Key values to report:

- ΔG TOTAL — total binding free energy (kcal/mol). More negative = stronger binding.
- ΔVDW AALS — van der Waals contribution (shape complementarity)
- ΔEEL — electrostatic contribution
- ΔEGB — desolvation penalty (always positive, opposes binding)

Note: MM-GBSA overestimates absolute ΔG values. Use for relative comparisons (mutant vs WT) rather than absolute affinity prediction.

Troubleshooting

Atom count mismatch

Error: 'The complex index has fewer atoms than the topology'

Cause: cpptraj and ParmEd stripped different residues. Fix: ensure both use identical strip commands including OPC.

Chain ID error

Error: 'No chain ID found' or 'list index out of range'

Fix: run pdb4amber again on system_amber.pdb. Check that system_final.pdb has chain IDs in column 22.

SOD atom type warning

Warning: 'Atom type SOD not present in AtomType list'

This is harmless. ParmEd still converts correctly. Ignore and proceed.